

Simulation results

massive-binary-sim v0.1.0 · git `UNKNOWN(no` · seed 42 · wallclock 0.12 s

1. Initial conditions

quantity	value
M1, M2 (ZAMS)	25, 18 Msun
separation / period	900 Rsun / 476.9 d
eccentricity	0
metallicity Z	0.0142
spins ($\Omega/\Omega_{\text{crit}}$)	0, 0

Active prescriptions

toggle	choice
gravity	`2.5PN`
wind	`vink2001`
convection	`ledoux`
overshoot	`exponential`
am_transport	`taylor_spruit`
mt_stability	`zeta`
mt_rate	`kolb_ritter`
ce_formalism	`alpha_lambda`
opacity	`kramers`
reaction_rates	`reaclib`
sn_engine	`delayed`
tides	`hut1981`
structure_fidelity	`semi_analytic`
third_body	`False`

2. Evolutionary timeline

t [Myr]	event	detail
0.022	star 1 -> MS	reduced-model phase
7.537	star 1 -> HG	reduced-model phase
7.674	star 1 -> CHeB	reduced-model phase
7.674	star 1 -> stripped_He	reduced-model phase
8.048	star 1 -> preSN	reduced-model phase
8.065	star 1 -> remnant	reduced-model phase
0.022	star 2 -> MS	reduced-model phase
8.304	star 2 -> HG	reduced-model phase
8.468	star 2 -> CHeB	reduced-model phase
8.468	star 2 -> stripped_He	reduced-model phase
8.891	star 2 -> preSN	reduced-model phase
8.891	star 2 -> remnant	reduced-model phase
7.674	**common_envelope**	survived: $a_f=47.179$ Rsun (spiral-in x19.5)
(at SN)	**SN1**	star 1 -> BH 3.44 Msun (delayed); bound after SN
8.468	**mass_transfer_end**	Case B transfer complete; donor -> 5.12 Msun stripped He star
(at SN)	**SN2**	star 2 -> NS 2.31 Msun (delayed); bound after SN
8.891	**dco_formed**	wide double compact binary (BH+NS), GW merger time 33921178.4 Gyr > Hubble

****Fate:**** wide double compact binary (BH+NS), GW merger time 33921178.4 Gyr > Hubble

3. Final-state table

quantity	star 1	star 2
mass [Msun]	3.443	2.308

phase remnant remnant
remnant BH 3.443 Msun (delayed)
remnant (2) NS 2.308 Msun
separation 1446 Rsun
period 2655 d
eccentricity 0.9109

4. Derived quantities

quantity value unit formula
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chirp mass M_c 2.444 Msun $M_c = (m_1 m_2)^{3/5} / (m_1+m_2)^{1/5}$
mass ratio q 0.6703 - $q = m_2 / m_1$ (≤ 1)
symmetric mass ratio η 0.2403 - $\eta = m_1 m_2 / (m_1+m_2)^2$
integrated wind loss (star1 / star2) 0.6197 Msun $\int \dot{M}_{\text{wind},1} dt$; star2 = 0.407 Msun
integrated RLOF transfer 13.44 Msun $\int \dot{M}_{\text{RLOF}} dt$ (gross, donor side)
net Delta M_1 21.56 Msun $M_1(\text{ZAMS}) - M_1(\text{final})$
net Delta M_2 -15.69 Msun $M_2(\text{final}) - M_2(\text{ZAMS})$
orbital separation (final) 1446 Rsun `state variable`
orbital period (final) 2655 day $P = 2 \pi \sqrt{a^3 / G(m_1+m_2)}$
orbital binding energy -1.0423e+46 erg $E = -G m_1 m_2 / (2a)$
orbital angular momentum 3.1412e+53 g cm ² / s $J = \mu \sqrt{G M a (1-e^2)}$
GW merger time 3.3921e+16 yr `Peters 1964 eq. 5.14 (eccentric)`
GW merger time 3.3921e+07 Gyr `as above`
peak GW frequency (ISCO) 764.6 Hz $f = c^3 / (6^{3/2} \pi G M)$
char. strain at ISCO (D=100 Mpc) 1.0873e-21 - `Maggiore 2007 eq. 4.44`

5. Error budget

****Numerical error**** (from the convergence study):

- not run in this invocation (`--convergence` flag)

****Physical / model uncertainty**** dominates and is quantified in §6-7. The reduced single-star evolution model (fitting formulae, not a structure)

6. Free-parameter sensitivity (one-at-a-time)

not run (`--sensitivity`)

7. Theory comparison (competing prescriptions)

not run (`--theory-compare`)

8. Validation suite results

test status detail
--- --- ---
kepler_closure PASS 200 orbits: max dE/E =1.77e-11, max dL/L =8.76e-12, da/a =1.52e-11 (tol 1e-06)
lane_emden_analytic PASS n=0: max dtheta =5.72e-16, dxil =8.88e-16; n=1: max dtheta =1.55e-14, dxil =1.82e-14; n=5: max dtheta =5.44e-11, dxil =1.82e-14
periastron_advance_1PN PASS 1PN d(omega)/orbit measured=6.157175e-05 rad, predicted 6 pi GM/(c^2 a(1-e^2))=6.155112e-05, frac diff=0.034%
hulse_taylor_decay PASS dPb/dt(Peters)=-2.402102e-12 s/s vs GR prediction -2.402630e-12 (Weisberg & Huang 2016); residual 0.022%. Merger
energy_conservation_2PN PASS 2PN conservative: secular dE/E over 120 orbits = 1.68e-08 (tol 1e-07); max dL/L =1.12e-05
eddington_limit_flag PASS 120 Msun ZAMS Gamma_Edd=0.406; 10x-luminosity state flags Gamma>1: True
solar_structure_eddington PASS Eddington model 1 Msun: P_c=1.256e+17 (ref 2.4e17), T_c=1.185e+07 (ref 1.57e7), rho_c=76.9 (ref 150), L=0.1

9. Assumptions & limitations

- ****1-D, orbit-averaged.**** No 3-D hydrodynamics; the orbit is treated by secular equations except in the dedicated Kepler-closure test.
- ****Reduced stellar evolution.**** Single-star tracks are fitting formulae (Hurley+ 2000-style), not Henyey structure integrations. Core masses, radii and lifetimes carry ~10-30% systematic uncertainty.

- ****No magnetic fields**** beyond the Tayler-Spruit AM-transport toggle.
- ****No rotational mixing feedback**** on the tracks (rotation only sets Ω_{crit} and tidal rates here).
- ****Mass-transfer rate**** uses a Ritter/Kolb-Ritter-style analytic nozzle with an efficiency prefactor (KNOWN_GAPS BIN-MDOT).
- ****GW waveform**** is a θ PN Newtonian chirp -- qualitative, not for matched filtering.
- ****3.5PN dynamics**** not implemented (KNOWN_GAPS PN-35).
- Full list: KNOWN_GAPS.md.

11. Bibliography

See `REFERENCES.bib` for the full list with DOIs. Key prescriptions used in this run: Peters (1964); Mora & Will (2004); Eggleton (1983); Ritte